Kepler’s laws

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use orbital shape, equal areas and period comparisons.

SUCCESS CRITERIA - I can...

1. Place the central body at a focus of an ellipse.
2. Compare speed near and far from the central body.
3. Use $T^2=a^3$ only with the specified units and central body.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Launch at **30 m/s** from level ground at **25°** or **65°**.
1. Use g=**10** and no drag.
2. **Compare ranges** and flight times.

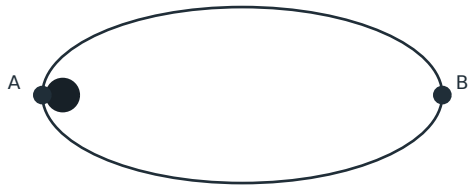
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $T^2 = a^3$.
For Sun orbits only when T is in years and a is in AU. a is the semi-major axis. Equal areas are swept in equal times.

Worked reasoning: For an ideal Sun orbit, a=**4 AU** gives $T=\sqrt{(\mathbf{4^3})}=$**8 years**. The orbit period depends on semimajor axis, not on a chosen point's instantaneous speed alone.

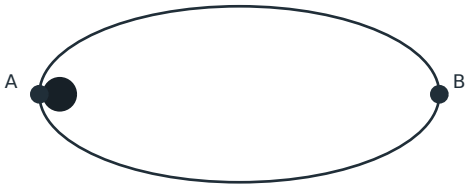
Your turn: Draw an ellipse, put the Sun at a focus and **mark the faster** and slower parts.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Mark the Sun at a focus on an ellipse. **Label where the object** is fastest and slowest.



Marzano check - circle: ① ② ③

04 Apply the physics

A Sun-orbit has a=**1.42 AU**. Compute period. Then **find period** at a=**4 AU** and **explain why doubling a** does not double T.

Semimajor axis (AU)	Period (years)
1.42	____
4	____

Marzano check - circle: ① ② ③

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Continue down each column.

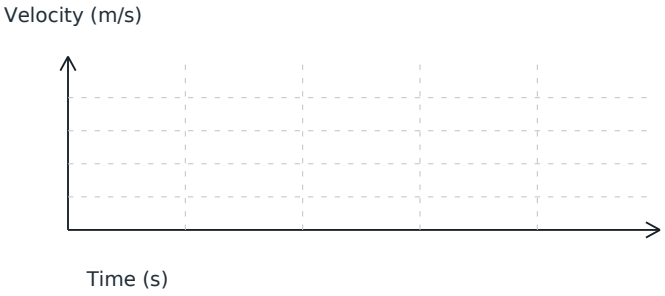
05 Explain your reasoning

Does knowing an orbit’s period alone tell the date when an asteroid reaches Earth’s location? **What else** is needed?

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

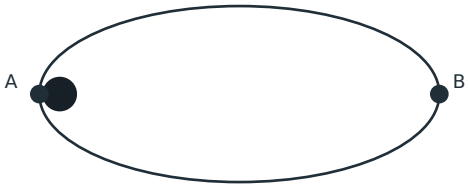
A ball begins with downward velocity **4 m/s** and falls **3 s**. Use downward positive, g=**10 m/s²** and no drag. **Find final velocity** and displacement.
Sketch the motion and **label a positive direction**.



Marzano check - circle: ① ② ③

07 Exit check

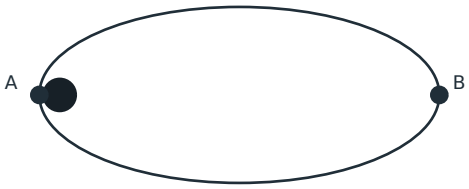
For two Sun-orbits, a_B=**2** a_A.
1. Find T_B/T_A.
2. **State why the same unit** form cannot automatically be used for a Moon orbit about Earth.



Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

An ideal Sun orbit has period **27 years**. **Find its semimajor axis**. If eccentricity is **0.5**, find perihelion and aphelion distances using r=a(**1**∓e).



Marzano check - circle: ① ② ③